

FIE401: Financial Econometrics

Instrumental Variables

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Basics of IV Regression

IV: Single Regressor and Single Instrument

IV: Multiple Regressors and Multiple Instruments

Checking Instrument Validity

Basics of IV Regression

IV Regression

- ▶ Three important threats to OLS validity are:
 - ▶ Omitted variable bias from a variable that is correlated with x but is unobserved (so cannot be included in the regression).
 - ▶ Simultaneous causality bias (x causes y , y causes x).
 - ▶ Errors-in-variables bias (x is measured with error).
- ▶ All three problems result in $E(u|x) \neq 0$. There is endogeneity problem!
- ▶ Instrumental variables regression can eliminate bias when $E(u|x) \neq 0$ – using an instrumental variable (IV), z .

IV: Single Regressor and Single Instrument

Two Conditions for a Valid Instrument

$$y = \beta_0 + \beta_1 x + u$$

- ▶ IV regression breaks x into two parts: a part that might be correlated with u , and a part that is not.
- ▶ By isolating the part that is not correlated with u , it is possible to estimate β_1 .
- ▶ This is done using an instrumental variable, z :
 - ▶ Relevant: $\text{Cov}(x, z) \neq 0$.
 - ▶ Exogenous: $\text{Cov}(z, u) = 0$.
- ▶ Suppose for now that you have such find such z . How can we use z to estimate β_1 ?

IV Estimator

$$y = \beta_0 + \beta_1 x + u$$

- ▶ x is endogenous (i.e., $E(u|x) \neq 0$) $\Rightarrow \text{Cov}(x, u) \neq 0$.
- ▶ z is exogenous (i.e., $E(u|z) = 0$) $\Rightarrow \text{Cov}(z, u) = 0$.

$$0 = \text{Cov}(z, u) = \text{Cov}(z, y - \beta_0 - \beta_1 x) = \text{Cov}(z, y) - \beta_1 \text{Cov}(z, x)$$

$$\beta_1 = \text{Cov}(z, y) / \text{Cov}(z, x).$$

IV Estimator: Two Stage Least Squares (2SLS) (i)

$$y = \beta_0 + \beta_1 x + u$$

First Stage:

- ▶ Isolate the part of x that is uncorrelated with u by regressing x on z using OLS.
- ▶ $x = \pi_0 + \pi_1 z + \nu$.
- ▶ Compute fitted values: $\hat{x} = \pi_0 + \pi_1 z$.
- ▶ Because z is uncorrelated with u , $\hat{x}$ is uncorrelated with u .

IV Estimator: Two Stage Least Squares (2SLS) (ii)

$$y = \beta_0 + \beta_1 x + u$$

Second Stage:

- ▶ Replace x by $\hat{x}$ in the regression of interest.
- ▶ $y = \beta_0 + \beta_1 \hat{x} + u$.
- ▶ Because $\hat{x}$ is uncorrelated with u , $E(u|\hat{x}) = 0$.
- ▶ This requires N to be large so that π_0 and π_1 are precisely estimated.
- ▶ The resulting estimator is called the Two Stage Least Squares (2SLS) estimator, β_1^{2SLS} and is consistent estimate of β_1 .
- ▶ The OLS standard errors from the second stage regression are not correct – they do not take into account the fact that $\hat{x}$ is an estimate from the first stage regression.

Example: Wage and Education

- ▶ What are returns to education for married women?
- ▶ If we just regress wage on the education itself we could have an omitted variable bias as we do not control for ability.
- ▶ Use father's education and/or mother's education to instrument for ability of the person.

Example: Wage and Education

inlf	hours	kidslt6	kidsge6	age	educ	wage	repwage
hushrs	husage	huseduc	huswage	faminc	mtr	motheduc	
fatheduc	unem	city	exper	nwifeinc	lwage	expersq	

Obs: 753

1. inlf	=1 if in labor force, 1975
2. hours	hours worked, 1975
3. kidslt6	# kids < 6 years
4. kidsge6	# kids 6-18
5. age	woman's age in yrs
6. educ	years of schooling
7. wage	estimated wage from earns., hours
8. repwage	reported wage at interview in 1976
9. hushrs	hours worked by husband, 1975
10. husage	husband's age
11. huseduc	husband's years of schooling
12. huswage	husband's hourly wage, 1975
13. faminc	family income, 1975
14. mtr	fed. marginal tax rate facing woman
15. motheduc	mother's years of schooling
16. fatheduc	father's years of schooling
17. unem	unem. rate in county of resid.
18. city	=1 if live in SMSA
19. exper	actual labor mkt exper
20. nwifeinc	(faminc - wage*hours)/1000
21. lwage	log(wage)
22. expersq	exper^2

Example: Wage and Education

```
#load necessary packages
require(AER);
#load data
load("M:/Projects/FIE401/data/data_lectures/WAGE.Rdata");
#Naive OLS
fit1<-lm(lwage~educ,data=WAGE);
#First stage
stage1<-lm(educ~fatheduc,data=WAGE);
#Fitted values
educ_hat<-fitted(stage1);
#Second stage
stage2<-lm(lwage~educ_hat,data=WAGE);
#First and second stage in one go
fit2<-ivreg(lwage~educ | fatheduc,data=WAGE);
```

Example: Wage and Education (Naive OLS)

```
##  
## Call:  
## lm(formula = lwage ~ educ, data = WAGE)  
##  
## Residuals:  
##      Min       1Q   Median       3Q      Max   
## -3.10256 -0.31473  0.06434  0.40081  2.10029   
##  
## Coefficients:  
##              Estimate Std. Error t value Pr(>|t|)      
## (Intercept)  -0.1852     0.1852  -1.000   0.318      
## educ          0.1086     0.0144   7.545 2.76e-13 ***  
## ---  
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1  
##  
## Residual standard error: 0.68 on 426 degrees of freedom  
## Multiple R-squared:  0.1179, Adjusted R-squared:  0.1158   
## F-statistic: 56.93 on 1 and 426 DF,  p-value: 2.761e-13
```

Example: Wage and Education (First Stage)

```
##
## Call:
## lm(formula = educ ~ fatheduc, data = WAGE)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -8.4704 -1.1231 -0.1231  0.9546  5.9546
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  10.23705    0.27594   37.099  <2e-16 ***
## fatheduc      0.26944    0.02859    9.426  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 2.081 on 426 degrees of freedom
## Multiple R-squared:  0.1726, Adjusted R-squared:  0.1706
## F-statistic: 88.84 on 1 and 426 DF,  p-value: < 2.2e-16
```

Example: Wage and Education (Second Stage)

```
##  
## Call:  
## lm(formula = lwage ~ educ_hat, data = WAGE)  
##  
## Residuals:  
##      Min       1Q   Median       3Q      Max   
## -3.2126 -0.3763  0.0563  0.4173  2.0604   
##  
## Coefficients:  
##              Estimate Std. Error t value Pr(>|t|)      
## (Intercept)  0.44110     0.46711   0.944   0.346      
## educ_hat     0.05917     0.03680   1.608   0.109      
##  
## Residual standard error: 0.7219 on 426 degrees of freedom  
## Multiple R-squared:  0.006034,    Adjusted R-squared:  0.003701   
## F-statistic: 2.586 on 1 and 426 DF,  p-value: 0.1086
```

Example: Wage and Education (IVREG)

```
##
## Call:
## ivreg(formula = lwage ~ educ | fatheduc, data = WAGE)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -3.0870 -0.3393  0.0525  0.4042  2.0677
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   0.44110    0.44610   0.989   0.3233
## educ          0.05917    0.03514   1.684   0.0929 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.6894 on 426 degrees of freedom
## Multiple R-Squared: 0.09344, Adjusted R-squared: 0.09131
## Wald test: 2.835 on 1 and 426 DF, p-value: 0.09294
```


IV: Multiple Regressors and Multiple Instruments

General IV

$$y = \beta_0 + \beta_1 x_1 + \dots + \beta_k x_k + \beta_{k+1} w_1 + \dots + \beta_{k+r} w_r + u$$

- ▶ So far we have considered IV regression with a single endogenous regressor (x) and a single instrument (z).
- ▶ We can easily extend this to:
 - ▶ multiple endogenous regressors ($x_1, \dots, x_k$).
 - ▶ multiple exogenous regressors ($w_1, \dots, w_r$) or control variables, which need to be included for the usual omitted variable reason.
 - ▶ multiple instrumental variables ($z_1, \dots, z_m$). More (relevant) instruments can produce a smaller variance of 2SLS: the R^2 of the first stage increases, so you have more variation in $\hat{x}$.

Identification

- ▶ Whether the coefficients are identified depends on the relation between the number of instruments (m) and the number of endogenous regressors (k).
- ▶ Intuitively, if there are fewer instruments than endogenous regressors, we cannot estimate $\beta_1, \dots, \beta_k$.
- ▶ For example, suppose $k = 1$ but $m = 0$ (no instruments)!
- ▶ The coefficients $\beta_1, \dots, \beta_k$ are said to be:
 - ▶ exactly identified if $m = k$. There are just enough instruments to estimate $\beta_1, \dots, \beta_k$.
 - ▶ overidentified if $m > k$. There are more than enough instruments to estimate $\beta_1, \dots, \beta_k$. If so, you can test whether the instruments are valid (a test of the “overidentifying restrictions”) - we will return to this later.
 - ▶ underidentified if $m < k$. There are too few instruments to estimate $\beta_1, \dots, \beta_k$. If so, you need to get more instruments!

IV estimator: 2SLS with a Single Endogenous Regressor

$$y = \beta_0 + \beta_1 x_1 + \beta_2 w_1 + \dots + \beta_{1+r} w_r + u$$

- ▶ m instruments.
- ▶ First Stage:
 - ▶ Regressing x_1 on z s and w s using OLS.
 - ▶ $x_1 = \pi_0 + \pi_1 z_1 + \dots + \pi_m z_m + \pi_{m+1} w_1 + \dots + \pi_{m+r} w_r + \nu$.
 - ▶ Compute fitted values: $\hat{x}_1$.
- ▶ Second Stage:
 - ▶ Replace x_1 by $\hat{x}_1$ in the regression of interest.
 - ▶ $y = \beta_0 + \beta_1 \hat{x}_1 + \beta_2 w_1 + \dots + \beta_{1+r} w_r + u$.
 - ▶ The resulting estimator is called the Two Stage Least Squares (2SLS) estimator, β_1^{2SLS} and is consistent estimate of β_1 (under usual assumptions).
 - ▶ In order to get correct standard errors, run 2SLS in one step.

Ws as control variables

- ▶ In many cases, the purpose of including the W s is to control for omitted factors, so that once the W s are included, z is uncorrelated with u .
- ▶ I.e., we are not interested in interpreting coefficient in front of W s.
- ▶ If so, W s do not need to be exogenous; instead, the W s need to be effective control variables – except now with a focus on producing an exogenous instrument.
- ▶ Technically, the condition for W s being effective control variables (instead of zero conditional mean):
$$E(u|w, z) = E(u|w).$$

Example: Wage and Education (General IV)

```
#load necessary packages
require(AER);
#load data
load("M:/Projects/FIE401/data/data_lectures/WAGE.Rdata");
#Naive OLS
fit1<-lm(lwage~educ+exper+I(exper^2),data=WAGE);
#First stage
stage1<-lm(educ~fatheduc+exper+I(exper^2),data=WAGE);
educ_hat<-fitted(stage1);
#Second stage
stage2<-lm(lwage~educ_hat+exper+I(exper^2),data=WAGE);
#First and second stage in one go
fit2<-ivreg(lwage~educ+exper+I(exper^2)
            | fatheduc+exper+I(exper^2),data=WAGE);
```

Example: Wage and Education (General IV)

```
##
## Call:
## ivreg(formula = lwage ~ educ + exper + I(exper^2) | fatheduc +
##       exper + I(exper^2), data = WAGE)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -3.09170 -0.32776  0.05006  0.37365  2.35346
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -0.0611169  0.4364461  -0.140  0.88870
## educ         0.0702263  0.0344427   2.039  0.04208 *
## exper        0.0436716  0.0134001   3.259  0.00121 **
## I(exper^2)   -0.0008822  0.0004009  -2.200  0.02832 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.6719 on 424 degrees of freedom
## Multiple R-Squared: 0.143,    Adjusted R-squared: 0.137
## Wald test: 8.314 on 3 and 424 DF,  p-value: 2.201e-05
```

Checking Instrument Validity

Checking Instrument Validity

- ▶ Relevance: **at least one** instrument must enter the population counterpart of the first stage regression.
- ▶ Exogenous: **all** the instruments must be uncorrelated with the error term.
 - ▶ What happens if one of these requirements is not satisfied?
 - ▶ How can we check?
 - ▶ If we have multiple instruments, which one should you use?

Relevance

$$y = \beta_0 + \beta_1 x_1 + \beta_2 w_1 + \dots + \beta_{1+r} w_r + u$$

First Stage:

- ▶ $x_1 = \pi_0 + \pi_1 z_1 + \dots + \pi_m z_m + \pi_{m+1} w_1 + \dots + \pi_{m+r} w_r + \nu$.
- ▶ The instruments are relevant if at least one of $\pi_1, \dots, \pi_m$ is non-zero.
- ▶ The instruments are said to be weak if all the $\pi_1, \dots, \pi_m$ are either zero or nearly zero.
- ▶ Weak instruments explain very little of the variation in x , beyond that explained by the w .
- ▶ Use F -statistic to test for weak instruments.

$$H_0: \pi_1 = \dots = \pi_m = 0$$

First-stage F -statistic

$$y = \beta_0 + \beta_1 x_1 + \beta_2 w_1 + \dots + \beta_{1+r} w_r + u$$

- ▶ Rule-of-thumb: If the first stage F -statistic is less than 10, then the set of instruments is weak.
- ▶ Why do we compare the first-stage F to 10?
- ▶ Simply rejecting the null hypothesis that the coefficients on the z are zero is not enough – we need substantial predictive content for the normal approximation to be a good one.
- ▶ Comparing the first-stage F to 10 tests whether the bias of 2SLS, relative to OLS, is less than 10%. If F is smaller than 10, the relative bias exceeds 10% – that is, 2SLS can have substantial bias.

Consequences of weak instruments

- ▶ Consider the simple case:
 - ▶ $y = \beta_0 + \beta_1 x + u$.
 - ▶ $x = \pi_0 + \pi_1 z + u$.
 - ▶ The IV estimator is $\beta_1^{2SLS} = \frac{\text{Cov}(y,z)}{\text{Cov}(x,z)}$.
- ▶ With weak instruments, the denominator is nearly zero.
- ▶ If instruments are weak, the sampling distribution of 2SLS and its t -statistics are not normal, even with N large.
- ▶ If instruments are weak, the usual methods of inference are unreliable.

Example: Wage and Education (Relevance)

```
#load necessary packages
require(lmtest);
#load data
load("M:/Projects/FIE401/data/data_lectures/WAGE.Rdata");
#First stage
stage1<-lm(educ~fatheduc+motheduc+exper+I(exper^2),
           data=WAGE);
#Define H0
myH0<-c("fatheduc=0", "motheduc=0");
#F-statistic
linearHypothesis(stage1,myH0);
```

Example: Wage and Education (Relevance)

```
## Linear hypothesis test
##
## Hypothesis:
## fatheduc = 0
## motheduc = 0
##
## Model 1: restricted model
## Model 2: educ ~ fatheduc + motheduc + exper + I(exper^2)
##
##   Res.Df    RSS Df Sum of Sq    F    Pr(>F)
## 1     425 2219.2
## 2     423 1758.6  2    460.64 55.4 < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Exogeneity (i)

- ▶ Exogenous: **all** the instruments must be uncorrelated with the error term.
- ▶ If the instruments are correlated with the error term, the first stage of 2SLS cannot isolate a component of x that is uncorrelated with the error term, so $\hat{x}$ is correlated with u and 2SLS is inconsistent.
- ▶ If there are more instruments than endogenous regressors, it is possible to test – partially – for instrument exogeneity.

Exogeneity (ii)

- ▶ Natural starting point is to compare the result of several 2SLS with a different set of instruments.
- ▶ But:
 - ▶ By eyeballing, we cannot say whether coefficients across several instrumental variable regressions are statistically different from each other.
 - ▶ More serious problem, is that coefficients might be similar even when all instruments are not exogenous (as far as all candidates for valid instruments are correlated with u in a similar manner).

Exogeneity (iii)

- ▶ Formal test (J -test):
 - ▶ Estimate 2SLS and obtain the residuals.
 - ▶ Regress the residuals on all exogenous variables: that is z and w .
$$\hat{u}^{2SLS} = \delta_0 + \delta_1 z_1 + \dots + \delta_m z_m + \delta_{m+1} w_1 + \dots + \delta_{m+r} w_r + \epsilon$$
 - ▶ Compute F-statistic for $H_0: \delta_1 = \dots = \delta_m = 0$.
 - ▶ $mF \sim \chi_q^2$ where $q = m - k$ under the H_0 : all instruments are exogenous.
- ▶ This can only be done in case $\# z > \# x$ (overidentified instrumental variable regression).
- ▶ If test rejects the H_0 , then z_1 or z_2 or both z_1 and z_2 must be invalid.
- ▶ The test does not tell us which is the right answer! You must exercise judgment!

Example: Wage and Education (Exogeneity)

```
#load necessary packages
require(AER);
require(lmtest);
#load data
load("M:/Projects/FIE401/data/data_lectures/WAGE.Rdata");
#IV regression
fit1<-ivreg(lwage~educ+exper+I(exper^2)
            | motheduc+fatheduc+exper+I(exper^2),
            data=WAGE);
WAGE$res<-residuals(fit1);
fit2<-lm(res~fatheduc+motheduc+exper+I(exper^2),
         data=WAGE);
#define H0
myH0<-c("fatheduc=0", "motheduc=0");
#test
#F-test
Fstat<-linearHypothesis(fit2,myH0)$F[2];
m<-2; #number of instruments
k<-1; #number of endogenous variables
testval<-m*Fstat;
pval<-1-pchisq(testval, (m-k));
```

Example: Wage and Education (Exogeneity)

```
## [1] "testval 0.37398464797992"
```

```
## [1] "pval 0.540840264706824"
```

Other issues

- ▶ All the logic can be extended for time-series and panel data as well.
- ▶ You might have to correct for heteroskedasticity / autocorrelation or cluster standard errors.
- ▶ IV panel regression in R:

```
ivreg(y~x+factor(entity) |  
      z+factor(entity),data=DF)
```

OR

```
plm(y~x | z,  
     model="within",data=DF)
```

Textbook

- ▶ Instrumental Variable: Chapter 12, Stock and Watson, Introduction to Econometrics, Global Edition, 4th edition